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PROJECT REPORT

ON

“AI POWERED DEEPPAKE FACE DETECTION ”

Submitted In Partial Fulfillment of The Requirements

For

VI SEMESTER

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY

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2024 – 25

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CERTIFICATE

This is to certify that the project entitled “**AI POWERED DEEPPAKE FACE DETECTION**” has been successfully submitted by **GANESH** RegNo. **U23IA22S0011**.

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor's degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

1.

2.

Guide
Name & Guide signature

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ACKNOWLEDGEMENT

I express my sincere gratitude to my guide, **Shri. Manjunath. V, Assistant Professor** of department of computer applications to guide for their guidance, proper advice & encouragement during the course or work.

I also feel very obliged to **Dr. Basavaprasad. B, Head of the Department** of computer applications for his encouragement & inspiration for execution of this thesis work.

I also feel thankful to **Dr. K Subhashchandra, Principal** of Government Degree College, Yadgir Support during work.

I am also thankful to the entire faculty & staff members and my friends from the computer applications department for their direct & indirect help & cooperation.

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CANDIDATE DECLARATION

Ganesh a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0011** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled "**Ai Powered Deepfake Face Detection**" submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of Bachelor's Degree In Computer Applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be slowly responsible for the same.

Date:-

Place:-

Signature of the candidate

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Abstract

Deepfake technology, which uses artificial intelligence (AI) to create hyper-realistic images and videos by manipulating faces, has become increasingly sophisticated. While this technology holds promise for entertainment and creative purposes, it also presents serious threats, such as the spread of misinformation, identity theft, and privacy violations. Detecting deepfake content has thus become crucial in maintaining trust in digital media.

The rapid advancement of deepfake technology has raised significant concerns regarding the authenticity of digital media, particularly in the context of face manipulation in videos and images. Deepfakes, which employ artificial intelligence (AI) to superimpose faces onto videos or images, have the potential to cause significant harm by spreading misinformation and undermining trust in media.

This project focuses on the development of an AI-powered system designed to detect deepfake faces in images and videos, leveraging advanced deep learning techniques. The system utilizes Convolutional Neural Networks (CNNs) for image classification, specifically trained to identify subtle inconsistencies in manipulated content, such as unnatural lighting, facial movements, and pixel-level artifacts that are characteristic of deepfakes.

For model training, publicly available datasets like FaceForensics++ and DeepFakeTIMIT were employed, providing diverse samples of both real and deepfake faces. The system was optimized to ensure high accuracy in detecting manipulated faces, achieving impressive results in distinguishing real from fake content.

Moreover, the system was tested to ensure real-time detection capabilities, enabling fast processing of video streams or images, which is essential for applications in areas like social media, security, and media verification. While the model performed effectively with high precision, recall, and accuracy, challenges remain in combating more sophisticated deepfake generation methods and ensuring efficient real-time deployment in resource-constrained environments.

This work presents a promising solution for combating the growing threat of deepfake media and ensures a more secure and reliable digital ecosystem.

Keywords: -

CNN, ResNet-50, DeepFake Detection, FaceForensics++, DeepFakeTIMIT Dataset, Transfer Learning, Pretrained Models, Image Manipulation, Real-Time Detection, TensorFlow, PyTorch, Binary Classification, Pixel-level Anomaly Detection, Data Augmentation, Model Evaluation, Precision, Recall, F1-Score, Confusion Matrix, False Positives, False Negatives, Image Preprocessing, Face Alignment, Adversarial Training, Cross-Validation, Anomaly Detection, Image Classification.

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